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### GUIDE WIRE AND MEDICAL SYSTEM

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#### Abstract

A guide wire includes: a conductive core shaft; a conductive coil body surrounding a distal end portion of the core shaft; a conductive distal tip joined to a distal end of the core shaft and a distal end of the coil body; and an insulating tube made of a resin and covering a proximal end portion of the coil body and a proximal end portion the core shaft. The coil body has: a covered portion covered by the insulating tube; and a protruding portion protruding from a distal end of the insulating tube toward the distal tip, and the protruding portion of the coil body and the distal tip constitute an electrode portion for plasma ablation.

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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This is a Continuation of Patent Cooperation Treaty Application Number PCT/JP2022/044157 filed Nov. 30, 2022. The disclosure of the prior application(s) is hereby incorporated by reference herein in its entirety.

### TECHNICAL FIELD

[0002] The disclosed embodiments relate to a guide wire and a medical system.

### BACKGROUND

[0003] As methods for treating arrhythmia resulting in abnormal heartbeat rhythm or chronic total occlusion (CTO) resulting from occlusion of a blood vessel by a lesion, plasma ablation therapies have been recently known, in which a living body tissue is ablated (cauterized) using a plasma flow. For example, Patent Literature 1 discloses a low-temperature plasma-type scalpel device. The device described in Patent Literature 1 includes a transmission electrode and a loop electrode that is inserted into a living body through the same catheter as of the transmission electrode, in which the transmission electrode generates a plasma to vaporize and excise a target by applying a voltage between the transmission electrode and the loop electrode.

### CITATION LIST

Patent Literature

[0004] Patent Literature 1: JP 2021-516138 W

### SUMMARY

Technical Problem

[0005] In the plasma ablation therapies, a plasma guide wire and a guide wire having a return electrode are inserted into a blood vessel, and, in this state, a high-frequency wave is applied to the plasma guide wire and the guide wire from a high-frequency generator. Thereby, a potential difference is generated between a distal end electrode provided on a distal end portion of the plasma guide wire and the return electrode provided on the guide wire, this potential difference causes a streamer corona discharge between the two electrodes. This streamer corona discharge makes it possible to ablate a CTO in the vicinity of the distal end electrode of the plasma guide wire. For correctly generating a plasma on the distal end electrode of the plasma guide wire, the return electrode provided on the guide wire needs to have a surface area larger than of the distal end electrode of the plasma guide wire.

[0006] Since a guide wire is inserted into a blood vessel, the guide wire needs to be configured to have a flexible distal end portion to improve safeness. However, in the device described in Patent Literature 1, improvement of the flexibility of the loop electrode on the distal end portion is not taken into account at all. Such problems are common to all guide wires to be inserted into living body lumens such as the lymph gland system, bile tract system, urinary system, respiratory tract system, digestive organ system, secreting gland system, and reproductive organs besides the blood vascular system. Furthermore, it has been required to improve the usability of the medical apparatus.

[0007] The disclosed embodiments have been made to solve at least a part of the above-described problems, and an object of the disclosed embodiments is to improve the flexibility of the distal end portion in the guide wire having an electrode portion on its distal end.

Solution to Problem

[0008] The disclosed embodiments have been made to solve at least a part of the above-described problems, and can be embodied as the following aspects. [0009] (1) According to an aspect of the

disclosed embodiments, a guide wire is provided. This guide wire includes a conductive core shaft, a conductive coil body surrounding a distal end portion of the core shaft, a conductive distal tip joined to a distal end of the core shaft and a distal end of the coil body, and an insulating tube made of a resin and covering a proximal end portion of the coil body and a proximal end portion of the core shaft. The coil body has a covered portion covered by the insulating tube, and a protruding portion protruding from a distal end of the insulating tube toward the distal tip. The protruding portion of the coil body and the distal tip constitute an electrode portion of the guide wire for plasma ablation.

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## Description

### BRIEF DESCRIPTION OF DRAWINGS

[0010] FIG. 1 is an explanatory view illustrating a configuration of a medical system.

[0011] FIG. 2 is an explanatory view illustrating a sectional configuration of a guide wire.

[0012] FIG. 3 is an enlarged sectional view illustrating a part of the guide wire on a distal end side.

[0013] FIG. 4 is an explanatory view explaining a method of using the medical system.

[0014] FIG. 5 is an enlarged sectional view illustrating a part of a guide wire on a distal end side, according to the second embodiment.

[0015] FIG. 6 is an enlarged sectional view illustrating a part of a guide wire on a distal end side, according to the third embodiment.

[0016] FIG. 7 is an enlarged sectional view illustrating a part of a guide wire on a distal end side, according to the fourth embodiment.

[0017] FIG. 8 is an explanatory view illustrating a sectional configuration of a guide wire according to the fifth embodiment.

[0018] FIG. 9 is an explanatory view illustrating a sectional configuration of a guide wire according to the sixth embodiment.

### DETAILED DESCRIPTION OF EMBODIMENTS

#### First Embodiment

[0019] FIG. 1 is an explanatory view illustrating a configuration of a medical system **1000**. The medical system **1000** is used for the purpose of recanalizing chronic total occlusion (CTO) or treating mild to moderate stenosis, significant stenosis, arrhythmia, or the like by ablating (cauterizing) a living body tissue using a plasma flow. The medical system **1000** includes a guide wire **1**, a plasma guide wire **100**, and a radio frequency (RF) generator **200**.

[0020] The plasma guide wire **100** is a device that has a distal end electrode DEL on its distal end, in which a plasma is generated on the distal end electrode DEL to ablate (cauterize) a target tissue such as CTO. The guide wire **1** is a device that has, on its distal end, an electrode portion EL that functionally serves as a return electrode and is used in combination with the plasma guide wire **100** to generate a plasma on the plasma guide wire **100**. The RF generator **200** is an apparatus that applies a high-frequency wave to the plasma guide wire **100** and the guide wire **1**. The disclosed embodiments will be explained below with reference to a case where the guide wire **1** and the plasma guide wire **100** are used for recanalizing CTO in a blood vessel, but the guide wire **1** and the plasma guide wire **100** may be inserted not only into vascular systems but also into living body lumens such as lymphatic systems, biliary systems, urinary systems, respiratory systems, digestive systems, secretory glands, and reproductive organs.

[0021] The plasma guide wire **100** has an elongated outer shape and includes a distal tip **180**, a first tube **110**, a second tube **120**, a third tube **130**, a core shaft **150**, a coil body **160**, a coil fixation portion **170**, a first fixation portion **171**, a second fixation portion **172**, a third fixation portion **173**, and a fourth fixation portion **174**.

[0022] The distal tip **180** is a conductive member that functionally serves as the distal end electrode

DEL for discharging electricity with the electrode portion EL of the guide wire **1**. The distal tip **180** is provided on the frontmost end side of the plasma guide wire **100**. The distal tip **180** has an outer shape with a diameter gradually decreasing from the proximal end side toward the distal end side for smoothening progress of the plasma guide wire **100** in a blood vessel and for facilitating generation of the plasma. As illustrated in FIG. **1**, the distal tip **180** according to the first embodiment has a shape closer to a triangular pyramid (triangular pyramid shape with a roundish tip) compared to a distal tip **80** of the guide wire **1**.

[0023] The core shaft **150** is a conductive member that constitutes a center axis of the plasma guide wire **100**. The core shaft **150** has an elongated outer shape extending in the longitudinal direction of the plasma guide wire **100**. The coil body **160** is conductive and surrounds a portion of the core shaft **150** on the distal end side. The coil body **160** is formed by spirally winding a conductive wire. The coil body **160** may be a single thread coil formed by winding one wire in a single thread manner; a multi-thread coil formed by winding a plurality of wires in a multiple thread manner; a single thread twisted wire coil formed by winding, in a single thread manner, a twisted wire with a plurality of wires twisted; or a multi-thread twisted wire coil formed by winding, in a multiple thread manner, each of a plurality of twisted wires with a plurality of wires twisted.

[0024] The first tube **110**, the second tube **120**, and the third tube **130** are all a hollow cylindrical tubular body made of an insulating resin. The first tube **110** is provided on the proximal end side with respect to the distal tip **180** to cover the distal end side of the core shaft **150** and the coil body **160**. The second tube **120** is provided on the proximal end side with respect to the third tube **130** to cover the proximal end side of the core shaft **150**. The third tube **130** is provided between the first tube **110** and the second tube **120** to cover an intermediate portion of the core shaft **150**. A distal end portion of the third tube **130** is joined to a proximal end portion of the first tube **110**. A proximal end portion of the third tube **130** is joined to a distal end portion of the second tube **120**. The third tube **130** has an outer diameter smaller than of the first tube **110** and smaller than of the second tube **120**. The third tube **130** has a distal end portion overlapped with the proximal end portion of the first tube **110**, and a proximal end portion overlapped with the distal end portion of the second tube **120**.

[0025] The first tube **110** has a gas layer **141** filled with a gas between the first tube **110** and the core shaft **150**/coil body **160**. The second tube **120** has a gas layer **142** filled with a gas between the second tube **120** and the core shaft **150**. The third tube **130** has a gas layer **143** filled with a gas between the third tube **130** and the core shaft **150**. As the gas constituting the gas layers **141**, **142**, and **143**, it is possible to use air, sulfur hexafluoride (SF.sub.6) gas, or hydrogen (H.sub.2) gas. When air is used as the gas, the gas layers **141**, **142**, and **143** may also be referred to as air layers **141**, **142**, and **143**, respectively.

[0026] The coil fixation portion **170** is a member that fixes the proximal end portion of the coil body **160** and a part of the core shaft **150**. The first fixation portion **171** is a member that is provided on the distal end portion of the first tube **110** to fix the distal end portion of the first tube **110**, the distal end portion of the core shaft **150**, and the distal end portion of the coil body **160**. The second fixation portion **172** is a member that is provided on the distal end portion of the third tube **130** to fix the distal end portion of the third tube **130**, the proximal end portion of the first tube **110**, and a part of the core shaft **150**. The third fixation portion **173** is a member that is provided on the proximal end portion of the third tube **130** to fix the proximal end portion of the third tube **130**, the distal end portion of the second tube **120**, and a part of the core shaft **150**. The fourth fixation portion **174** is a member that is provided on the proximal end portion of the second tube **120** to fix the proximal end portion of the second tube **120** and the proximal end portion of the core shaft **150**.

[0027] The core shaft **150** and the distal tip **180** can be made of any conductive material, such as a chromium-molybdenum steel, a nickel-chromium-molybdenum steel, a stainless steel such as SUS304, and a nickel-titanium alloy. The distal tip **80** may be formed by melting the distal end portion of the core shaft **150** with a laser or the like. The first tube **110**, the second tube **120**, and

the third tube **130** can be made of any insulating material, e.g. a copolymer of tetrafluoroethylene and perfluoroalkoxy ethylene (PFA); a polyolefin such as polyethylene, polypropylene, and ethylene-propylene copolymer; a polyester such as polyethylene terephthalate; polyvinyl chloride; an ethylene-vinyl acetate copolymer; a crosslinked ethylene-vinyl acetate copolymer; a thermoplastic resin such as polyurethane; a polyamide elastomer; a polyolefin elastomer; a silicone rubber; a latex rubber; and a super engineering plastic such as polyetheretherketone, polyetherimide, polyamide-imide, polysulfone, polyimide, and polyethersulfone. Each of the first tube **110**, the second tube **120**, and the third tube **130** may be made of a same material, or may be made of different materials. The coil fixation portion **170**, the first fixation portion **171**, the second fixation portion **172**, the third fixation portion **173**, and the fourth fixation portion **174** may be made of any bonding agent such as an epoxy adhesive.

[0028] The RF generator **200** outputs a high frequency power to between a first terminal **210** and a second terminal **220**. From the first terminal **210**, a first cable **211** extends. The first cable **211** is connected to a proximal end portion **155** of the plasma guide wire **100**. From the second terminal **220**, a second cable **221** extends. The second cable **221** is connected to the proximal end portion **55** of the guide wire **1**. The first cable **211** and the second cable **221** are conductive electric wires. The first cable **211** and the second cable **221** may have a cable connector (connection terminal for physical and electrical connection between cables). The RF generator **200** functionally serves as a “high-frequency generator”.

[0029] FIG. **2** is an explanatory view illustrating a sectional configuration of the guide wire **1**. FIG. **3** is an enlarged sectional view illustrating a part of the guide wire **1** on the distal end side. FIG. **2** and FIG. **3** are sectional views taken along the longitudinal section, as in FIG. **1**. The configuration of the guide wire **1** will be explained below with reference to FIG. **1** to FIG. **3**. As described above, the guide wire **1** has, on its distal end, the electrode portion **EL** that functionally serves as a return electrode, and used in combination with the plasma guide wire **100** to generate a plasma on the plasma guide wire **100**.

[0030] In FIG. **2** and FIG. **3**, an axis passing through the center of the guide wire **1** is expressed by an axis line **O** (dashed-dotted line). In the examples of FIG. **2** and FIG. **3**, the axis line **O** coincides with the axis passing through the center of each constituent member of the guide wire **1**, i.e. a first tube **10**, a second tube **20**, a third tube **30**, the distal tip **80**, a core shaft **50**, and a coil body **60**. However, the axis line **O** may be inconsistent with the center axis of each constituent member in the guide wire **1**. In FIG. **2** and FIG. **3**, XYZ axes orthogonal to each other are illustrated. The X axis corresponds to the longitudinal direction of the guide wire **1**, the Y axis corresponds to the height direction of the guide wire **1**, and the Z axis corresponds to the width direction of the guide wire **1**. The left side (−X axis direction) in FIG. **2** and FIG. **3** is referred to as “distal end side” of the guide wire **1** and each constituent member, and the right side (+X axis direction) in FIG. **2** and FIG. **3** is referred to as “proximal end side” of the guide wire **1** and each constituent member. Of both ends in the longitudinal direction (X axis direction), one end located on the distal end side is referred to as “distal end”, and the other end located on the proximal end side is referred to as “proximal end”. The distal end and its vicinity are referred to as “distal end portion”, and the proximal end and its vicinity are referred to as “proximal end portion”. The distal end side is inserted into a living body, and the proximal end side is operated by an operator such as a surgeon. The same applies to the figures following FIG. **2** and FIG. **3**.

[0031] The guide wire **1** has an elongated outer shape and includes the first tube **10**, the second tube **20**, the third tube **30**, the core shaft **50**, the distal tip **80**, the coil body **60**, a coil fixation portion **70**, a first fixation portion **71**, a second fixation portion **72**, a third fixation portion **73**, a fourth fixation portion **74**.

[0032] The distal tip **80** is conductive member that discharges electricity on the distal tip **180** of the plasma guide wire **100** used in combination with the guide wire **1** (FIG. **1**). The distal tip **80** is provided on the frontmost end side of the guide wire **1** (i.e. the distal end portion of the guide wire

1). The distal tip **80** has an outer shape with a diameter gradually decreasing from the proximal end side toward the distal end side for smoothening progress of the guide wire **1** in a blood vessel. As illustrated in FIG. 2 and FIG. 3, the distal tip **80** according to the first embodiment is hemispherical. The maximum outer diameter of the distal tip **80** (i.e. the outer diameter of the proximal end portion of the distal tip **80**) is substantially equal to an outer diameter  $\Phi 1$  of a protruding portion **61** described later. The proximal end portion of the distal tip **80** is joined to the distal end of the core shaft **50** and a distal end **68** of the coil body **60**. Any bonding agent such as an epoxy adhesive can be used for the joining. Further, laser welding or the like may be used as a joining means.

[0033] The core shaft **50** is a conductive member that constitutes the center axis of the guide wire **1**. The core shaft **50** has an elongated outer shape extending in the longitudinal direction of the guide wire **1**. The core shaft **50** includes a small diameter portion **51**, a first tapered portion **52**, a second tapered portion **53**, and a large diameter portion **54** in this order from the distal end toward the proximal end. The small diameter portion **51** is a portion where the outer diameter of the core shaft **50** is the smallest, and has a substantially columnar shape with a constant outer diameter from the distal end to the proximal end. The first tapered portion **52** is provided between the small diameter portion **51** and the second tapered portion **53** and has an outer shape with a diameter gradually decreasing from the proximal end side toward the distal end side. The second tapered portion **53** is provided between the first tapered portion **52** and the large diameter portion **54** and has an outer shape with an outer diameter gradually decreasing from the proximal end side toward the distal end side at an inclination angle different from that of the first tapered portion **52**. The large diameter portion **54** is a portion where the outer diameter of the core shaft **50** is the largest and has a substantially columnar shape with a constant outer diameter from the distal end to the proximal end. The proximal end portion **55** of the large diameter portion **54** is a portion where a proximal end surface of the large diameter portion **54** bulges.

[0034] In the first embodiment, the “constant” is synonymous with “substantially constant”, and means that the diameter is substantially constant while accepting fluctuation due to production errors or the like. In the first embodiment, when the transverse section of the member (or the inner cavity) is elliptical, the “outer diameter” and the “inner diameter” refer to the length of the largest diameter in any transverse section.

[0035] The coil body **60** is conductive and arranged so as to surround a part of the core shaft **50** on the distal end side. In the example of FIG. 2, the coil body **60** surrounds, in the core shaft **50**, the small diameter portion **51** and a part of the first tapered portion **52** on the distal end side. The coil body **60** is formed by spirally winding a conductive wire. The coil body **60** may be a single thread coil formed by winding one wire in a single thread manner; a multi-thread coil formed by winding a plurality of wires in a multiple thread manner; a single thread twisted wire coil formed by winding, in a single thread manner, a twisted wire with a plurality of wires twisted; or a multi-thread twisted wire coil formed by winding, in a multiple thread manner, each of a plurality of twisted wires with a plurality of wires twisted.

[0036] As illustrated in FIG. 2, the coil body **60** has a protruding portion **61** and a covered portion **62**. The covered portion **62** is a portion of the coil body **60**, which is covered by the insulating tubes **10**, **20**, and **30** (specifically, the first tube **10**). The protruding portion **61** is a portion of the coil body **60**, which is not covered by the insulating tubes **10**, **20**, and **30** but protrudes from a distal end **11a** of the insulating tubes **10**, **20**, and **30** (specifically, first tube **10**) toward the distal tip **80** side. In other words, it can be said that the protruding portion **61** is a portion of the coil body **60**, which protrudes from the distal end **11a** of the first tube **10** toward the distal end side (in the  $-X$  axis direction).

[0037] As illustrated in FIG. 3, the protruding portion **61** of the coil body **60** has a straight portion **611** and a tapered portion **612**. The straight portion **611** is a portion of the protruding portion **61**, which has a constant outer diameter  $\Phi 1$ . The straight portion **611** is provided on the frontmost end

side of the coil body **60** (i.e. on the distal end side with respect to the tapered portion **612**). The tapered portion **612** is a portion of the protruding portion **61**, which is provided on the proximal end side with respect to the straight portion **611** and has an outer diameter gradually decreasing from the distal end toward the proximal end. In the first embodiment, the “outer diameter” of the protruding portion **61**, the covered portion **62**, the straight portion **611**, and the tapered portion **612** refers to the outer diameter of the thickest portion of the wire constituting each portion.

[0038] The first tube **10**, the second tube **20**, and the third tube **30** are all hollow cylindrical tubular bodies made of an insulating resin. The first tube **10**, the second tube **20**, and the third tube **30** are also collectively referred to as “insulating tubes **10**, **20**, and **30**”.

[0039] The first tube **10** is provided on the proximal end side with respect to the distal tip **80** and the protruding portion **61** and on the distal end side with respect to the second tube **20** and the third tube **30**. The first tube **10** covers the covered portion **62** of the coil body **60** and a part of the core shaft **50** (specifically, a part of the small diameter portion **51** on the proximal end side and a part of the first tapered portion **52** on the distal end side). The inner diameter of the first tube **10** is larger than the outer diameter of the covered portion **62**. The thickness and length of the first tube **10** may be arbitrarily determined. The second tube **20** is provided on the proximal end side with respect to the first tube **10** and the second tube **20**. The second tube **20** covers the proximal end portion of the first tapered portion **52** of the core shaft **50**, the second tapered portion **53**, and the large diameter portion **54**. The proximal end portion **55** of the large diameter portion **54** is not covered by the second tube **20** but is exposed to the outside. The second tube **20** has an inner diameter larger than the outer diameter of the large diameter portion **54** of the core shaft **50**. A thickness and a length of the second tube **20** may be arbitrarily determined.

[0040] The third tube **30** is provided between the first tube **10** and the second tube **20**. The third tube **30** covers an intermediate portion of the first tapered portion **52** of the core shaft **50**. A thickness and a length of the third tube **30** may be arbitrarily determined. As illustrated in FIG. 2, the distal end portion **31** of the third tube **30** is joined to a proximal end portion **12** of the first tube **10**. A proximal end portion **32** of the third tube **30** is joined to a distal end portion **21** of the second tube **20**. The third tube **30** has an outer diameter smaller than of the first tube **10** and smaller than of the second tube **20**. Further, as illustrated in FIG. 2, the third tube **30** is arranged such that the distal end portion **31** of the third tube **30** is overlapped with the proximal end portion **12** of the first tube **10** and the proximal end portion **32** of the third tube **30** is overlapped with the distal end portion **21** of the second tube **20**. Thus, in the third tube **30**, an outer peripheral surface **34** of the distal end portion **31** is joined to an inner peripheral surface **13** of the proximal end portion **12** of the first tube **10** and the outer peripheral surface **34** of the proximal end portion **32** is joined to an inner peripheral surface **23** of the distal end portion **21** of the second tube **20**. A portion of the third tube **30**, which is located between the distal end portion **31** and the proximal end portion **32** (intermediate portion) is not covered with the first tube **10** or the second tube **20** but is exposed to the outside.

[0041] For the joining of the first tube **10**, the second tube **20**, and the third tube **30**, any bonding agent such as an epoxy adhesive can be used. In FIG. 2, the joint portion between the third tube **30** and the first tube **10** is illustrated as a distal end side joint part **82** (dashed circle frame), and the joint portion between the third tube **30** and the second tube **20** is illustrated as a proximal end side joint part **83** (dashed circle frame). As illustrated in FIG. 2, the insulating tubes **10**, **20**, and **30** according to the first embodiment have a constricted shape on the intermediate portion where the third tube **30** is provided.

[0042] As illustrated in FIG. 2, the first tube **10** has a gas layer **41** filled with a gas between the inner peripheral surface **13** of the first tube **10** and the outer peripheral surfaces of the core shaft **50**/covered portion **62**. The gas layer **41** is provided entirely in the circumferential direction. The gas layer **41** is provided entirely in the longitudinal direction from the distal end portion **11** to the proximal end portion **12** of the first tube **10** (specifically, entirely in the longitudinal direction from

the proximal end of the first fixation portion **71** to the distal end of the second fixation portion **72**) excluding sites where the first fixation portion **71** and the second fixation portion **72** are provided. The second tube **20** has a gas layer **42** filled with a gas between the inner peripheral surface **23** of the second tube **20** and the outer peripheral surface of the core shaft **50**. Similarly to the gas layer **41**, the gas layer **42** is provided entirely in the circumferential direction. The gas layer **42** is provided entirely in the longitudinal direction from the distal end portion **21** to the proximal end portion **22** of the second tube **20** (specifically, entirely in the longitudinal direction from the proximal end of the third fixation portion **73** to the distal end of the fourth fixation portion **74**) excluding sites where the third fixation portion **73** and the fourth fixation portion **74** are provided. The third tube **30** has a gas layer **43** filled with a gas between an inner peripheral surface **33** of the third tube **30** and the outer peripheral surface of the core shaft **50**. Similarly to the gas layer **41**, the gas layer **43** is provided entirely in the circumferential direction. The gas layer **43** is provided entirely in the longitudinal direction from the distal end portion **31** to the proximal end portion **32** of the third tube **30** (specifically, entirely in the longitudinal direction from the proximal end of the second fixation portion **72** to the distal end of the third fixation portion **73**) excluding sites where the second fixation portion **72** and the third fixation portion **73** are provided.

[0043] As the gas constituting the gas layers **41**, **42**, and **43**, any gas may be used as long as the gas is more electrically insulative than the insulating resin constituting the first, second, and third tubes **10**, **20**, and **30**. As the gas constituting the gas layers **41**, **42**, and **43**, for example, air, sulfur hexafluoride (SF<sub>6</sub>) gas, or hydrogen (H<sub>2</sub>) gas may be used. When air is used as the gas, the gas layers **41**, **42**, and **43** may also be referred to as air layers **41**, **42**, and **43**.

[0044] The coil fixation portion **70** fixes a proximal end **69** of the covered portion **62** of the coil body **60** and a part of the first tapered portion **52** of the core shaft **50**. The first fixation portion **71** is provided on the distal end portion **11** of the first tube **10** to fix the distal end portion **11** of the first tube **10**, a part of the coil body **60** (specifically, a boundary portion between the protruding portion **61** and the covered portion **62**), and a part of the small diameter portion **51** of the core shaft **50**. The first fixation portion **71** is provided entirely in the circumferential direction to block the gas flow inside and outside of the guide wire **1** (specifically, the flow of the gas constituting the gas layer **41**).

[0045] The second fixation portion **72** is provided on the distal end portion **31** of the third tube **30** to fix the distal end portion **31** of the third tube **30**, the proximal end portion **12** of the first tube **10**, and a part of the first tapered portion **52** of the core shaft **50**. The second fixation portion **72** is provided entirely in the circumferential direction to block the gas flow between the gas layer **41** and the gas layer **43**. The third fixation portion **73** is provided on the proximal end portion **32** of the third tube **30** to fix the proximal end portion **32** of the third tube **30**, the distal end portion **21** of the second tube **20**, and a part of the first tapered portion **52** of the core shaft **50**. The third fixation portion **73** is provided entirely in the circumferential direction to block the gas flow between the gas layer **43** and the gas layer **42**. The fourth fixation portion **74** is provided on the proximal end portion **22** of the second tube **20** to fix the proximal end portion **22** of the second tube **20** and the proximal end portion of the large diameter portion **54** of the core shaft **50**. The fourth fixation portion **74** is provided entirely in the circumferential direction to block the gas flow inside and outside of the guide wire **1** (specifically, the flow of the gas constituting the gas layer **42**).

[0046] The core shaft **50** and the distal tip **80** can be made of any conductive material, such as a chromium-molybdenum steel, a nickel-chromium-molybdenum steel, a stainless steel such as SUS304, and a nickel-titanium alloy. The distal tip **80** may be formed by melting the distal end portion of the core shaft **50** with a laser or the like.

[0047] The first tube **10**, the second tube **20**, and the third tube **30** can be made of any insulating material, e.g. a copolymer of tetrafluoroethylene and perfluoroalkoxy ethylene (PFA); a polyolefin such as polyethylene, polypropylene, and ethylene-propylene copolymer; a polyester such as polyethylene terephthalate; polyvinyl chloride; an ethylene-vinyl acetate copolymer; a crosslinked



ethylene-vinyl acetate copolymer; a thermoplastic resin such as polyurethane; a polyamide elastomer; a polyolefin elastomer; a silicone rubber; a latex rubber; and a super engineering plastic such as polyetheretherketone, polyetherimide, polyamide-imide, polysulfone, polyimide, and polyethersulfone. Each of the first tube **10**, the second tube **20**, and the third tube **30** may be made of a same material, or may be made of different materials depending on the performance required for the guide wire **1** (e.g. flexibility, torquability, and shape maintainability of the distal end portion).

[0048] The coil fixation portion **70**, the first fixation portion **71**, the second fixation portion **72**, the third fixation portion **73**, and the fourth fixation portion **74** can be made of any bonding agent such as an epoxy adhesive.

[0049] As described above, the coil body **60** and the distal tip **80** are electrically conductive, and meanwhile the insulating tubes **10**, **20**, and **30** are electrically insulative. As illustrated in FIG. 3, in the guide wire **1** according to the first embodiment, the protruding portion **61** of the coil body **60** protrudes from the distal end **11a** of the insulating tube **10**, **20**, and **30** (specifically, the first tube **10**) toward the distal tip **80** side. As a result, the protruding portion **61** is not covered by the insulating tubes **10**, **20**, and **30** but exposed to the outside, and therefore can functionally serve as the electrode portion EL together with the distal tip **80**. The electrode portion EL is used as a return electrode for the distal end electrode DEL (FIG. 1) of the plasma guide wire **100**, i.e. an electrode for plasma ablation. The surface of the wire constituting the protruding portion **61** may be coated with a conductive resin or the like. Also in this case, the protruding portion **61** is not covered by the insulating tubes **10**, **20**, and **30** but is exposed, and therefore can functionally serve as the electrode portion EL.

[0050] For correctly generating a plasma on the distal end electrode DEL of the plasma guide wire **100**, the surface area of the electrode portion EL provided on the guide wire **1** needs to be larger than the surface area of the distal end electrode DEL of the plasma guide wire **100**. In this regard, according to the configuration explained in FIG. 2 and FIG. 3, the surface area of the electrode portion EL is a sum of the surface area of the protruding portion **61** and the surface area of the distal tip **80**. In other words, the surface area of the protruding portion **61** can be added to the surface area of the electrode portion EL.

[0051] As illustrated in FIG. 3, since the straight portion **611** of the protruding portion **61** has a constant outer diameter  $\Phi 1$ , the outer diameter of the proximal end of the straight portion **611** is also  $\Phi 1$ . In the guide wire **1** according to the first embodiment, the outer diameter  $\Phi 1$  of the proximal end of the straight portion **611** is equal to the outer diameter  $\Phi 10$  of the distal end of the insulating tubes **10**, **20**, and **30** (specifically, outer diameter  $\Phi 10$  of the distal end **11a** of the first tube **10**). Herein, the terms “the same” and “equal” include not only a meaning of strict conformity but also a meaning of allowing differences due to production errors or the like. In FIG. 3, for convenience of illustration, the arrow indicating the outer diameter  $\Phi 10$  is illustrated slightly on the proximal end side with respect to the distal end **11a** of the first tube **10**.

[0052] FIG. 4 is a diagram illustrating the method of using the medical system **1000**. First, an operator inserts a delivery guide wire into a blood vessel **400** and delivers the delivery guide wire to the vicinity of a CTO**401**. The operator inserts the delivery guide wire into a catheter **300** and delivers the catheter **300** along the delivery guide wire to the vicinity of the CTO**401**. In FIG. 4, the catheter **300** is exemplified by a so-called multilumen catheter including a first shaft **301** having a first lumen **301L**, a second shaft **302** having a second lumen **302L**, and a distal tip **303**. The operator inserts the plasma guide wire **100** into the first lumen **301L** and protrudes the distal end electrode DEL of the plasma guide wire **100** to the outside from a distal end opening of the first lumen **301L** to place the distal end electrode DEL on the vicinity of a tissue to be ablated (CTO**401**). Similarly, the operator inserts the guide wire **1** into the second shaft **302** and protrudes the electrode portion EL of the guide wire **1** to the outside from the distal end opening of the second lumen **302L**. In this state, the operator outputs a high-frequency power from the RF

generator **200**. Then, due to a potential difference between the distal end electrode DEL of the plasma guide wire **100** and the electrode portion EL of the guide wire **1**, streamer corona discharge occurs between the distal end electrode DEL and the electrode portion EL. This streamer corona discharge makes it possible to ablate the CTO in the vicinity of the distal end electrode DEL of the plasma guide wire **100**, as illustrated in FIG. 4.

[0053] Although a case where the guide wire **1** and the plasma guide wire **100** is delivered using the catheter **300** as the multilumen catheter has been described as an example in FIG. 4, the guide wire **1** and the plasma guide wire **100** may be delivered without using the catheter **300**. The guide wire **1** and the plasma guide wire **100** may be individually delivered using two different catheters. Although a case where another delivery guide wire is used for delivering the catheter **300** has been described as an example in FIG. 4, the guide wire **1** or the plasma guide wire **100** may be used as a delivery guide wire. Although a case where the CTO **401** is ablated from a true lumen of the blood vessel **400** has been described as an example in FIG. 4, it is possible to adopt a process in which a guide wire is once proceeded from the true lumen of the blood vessel **400** to a false lumen, then the end of the CTO **401** is ablated from the false lumen, and the guide wire is proceeded to the true lumen. In this case, it is possible to place both the guide wire **1** and the plasma guide wire **100** in the false lumen, or alternatively it is possible to place only the plasma guide wire **100** in the false lumen and to place the guide wire **1** in the true lumen. Furthermore, although a case where the distal end electrode DEL of the plasma guide wire **100** is located on the distal end side with respect to the electrode portion EL in the blood vessel **400** has been described as an example in FIG. 4, this positional relationship between the two electrodes may be reversed, or the two electrodes may be at the same position. As described above, the guide wire **1** and the plasma guide wire **100** can be used in any way and in combination of any devices.

[0054] As described above, in the guide wire **1** according to the first embodiment, the coil body **60** has the protruding portion **61** that is not covered by the insulating tubes **10**, **20**, and **30** but protrudes from the distal end **11a** of the insulating tubes **10**, **20**, and **30** toward the distal tip **80** side, in which the protruding portion **61** of the coil body **60** and the distal tip **80** constitute the electrode portion EL for plasma ablation. That means, in the guide wire **1** according to the first embodiment, since the surface area of the electrode portion EL is a sum of the surface area of the protruding portion **61** of the coil body **60** and the surface area of the distal tip **80** (i.e. the surface area of the protruding portion **61** of the coil body **60** can be added to the surface area of the electrode portion EL), the surface area of the electrode portion EL can be easily increased compared to the conventional configurations with an electrode portion consisting only of a distal tip. If the electrode portion consists only of the distal tip, the distal tip needs to be enlarged for increasing the surface area of the electrode portion, which may compromise the flexibility of the distal end portion of the guide wire. However, in the guide wire **1** according to the first embodiment, since the surface area of the electrode portion EL is a sum of the surface area of the protruding portion **61** of the coil body **60** and the surface area of the distal tip **80**, the distal tip **80** need not be excessively enlarged, and there is no risk of compromising the flexibility of the distal end portion in the guide wire **1**. Furthermore, the protruding portion **61** has a coil shape, and therefore, even if the distal end portion (electrode portion EL) of the guide wire **1** collides with a blood vessel wall, the impact can be relieved by the protruding portion **61**, and damage to the blood vessel wall can be suppressed. As a result, in the guide wire **1** having the electrode portion EL on its distal end, the flexibility of the distal end portion and the safeness of the procedure can be improved.

[0055] In the guide wire **1** according to the first embodiment, since the protruding portion **61** of the coil body **60** has the straight portion **611** having a constant outer diameter  $\Phi 1$ , and the tapered portion **612** provided on the proximal end side with respect to the straight portion **611** and having an outer diameter gradually decreasing from the distal end to the proximal end, the surface area of the protruding portion **61** can be increased while the outer diameter  $\Phi 1$  of the protruding portion **61** is maintained constant, i.e. the surface area of the electrode portion EL can be increased. Since the

large surface area of the electrode portion EL can contribute not only to correct plasma generation on the distal end electrode DEL of the plasma guide wire **100** used in combination with the guide wire **1** but also to improvement in visibility of the electrode portion EL in an X-ray image (angiogram), the usability of the guide wire **1** can be improved. Also, the increase in the surface area of the electrode portion EL leads to reduction in the risk of blood vessel perforation, and therefore the safeness of the guide wire **1** can be improved. Since the protruding portion **61** of the coil body **60** has, on the proximal end side with respect to the straight portion **611**, the tapered portion **612** having the outer diameter gradually decreasing from the distal end toward the proximal end, a rigidity of the protruding portion **61** can be gradually changed, and as a result, fracture of the protruding portion **61** due to a rigidity gap can be suppressed.

[0056] Furthermore, in the guide wire **1** according to the first embodiment, since the outer diameter  $\Phi 1$  of the proximal end of the straight portion **611** is equal to the outer diameters  $\Phi 10$  of the distal ends of the insulating tubes **10**, **20**, and **30** (specifically, first tube **10**), the outer diameter of the entire distal end side (specifically, the electrode portion EL excluding the tapered portion **612**, and the first tube **10**) of the guide wire **1** can be made constant. As a result, the guide wire **1** can be prevented from being caught in the blood vessel or caught by other devices (e.g. plasma guide wire **100**, and catheter **300** in FIG. 4).

[0057] Furthermore, in the guide wire **1** according to the first embodiment, since the electrode portion EL is a return electrode, the guide wire **1** can be configured as a so-called return guide wire that is used in combination with the plasma guide wire **100**.

### Second Embodiment

[0058] FIG. 5 is an enlarged sectional view illustrating a part of a guide wire **1A** on a distal end side, according to the second embodiment. The guide wire **1A** according to the second embodiment includes a coil body **60A** instead of the coil body **60** and a distal tip **80A** instead of the distal tip **80** in the configuration described in the first embodiment.

[0059] The coil body **60A** includes a protruding portion **61A** instead of the protruding portion **61**. The protruding portion **61A** does not include the straight portion **611** and the tapered portion **612** described in the first embodiment, and has a straight shape with an outer diameter  $\Phi 1A$  constant in whole. The outer diameter  $\Phi 1A$  of the protruding portion **61A** is equal to the outer diameter of the covered portion **62**. That means, the coil body **60A** has a constant outer diameter from the distal end to the proximal end. The outer diameter  $\Phi 1A$  of the protruding portion **61A** is smaller than the outer diameter  $\Phi 10$  of the distal ends of the insulating tubes **10**, **20**, and **30** (specifically, the outer diameter  $\Phi 10$  of the distal end **11a** of the first tube **10**). The maximum outer diameter of the distal tip **80A** (i.e. the outer diameter of the proximal end portion of the distal tip **80A**) is substantially equal to the outer diameter  $\Phi 1A$  of the protruding portion **61A**.

[0060] As described above, the configuration of the coil body **60A** can be variously modified, and the coil body **60A** may be configured so as to have a constant outer diameter from the distal end to the proximal end. Also, the guide wire **1A** according to the second embodiment as described above can exhibit the same effect as in the aforementioned first embodiment. According to the configuration of the second embodiment, the process for the guide wire **1** can be simplified.

### Third Embodiment

[0061] FIG. 6 is an enlarged sectional view illustrating a part of a guide wire **1B** on a distal end side, according to the third embodiment. The guide wire **1B** according to the third embodiment includes a coil body **60B** instead of the coil body **60** and a first fixation portion **71B** instead of the first fixation portion **71** in the configuration described in the first embodiment.

[0062] The coil body **60B** includes a protruding portion **61B** instead of the protruding portion **61** and a covered portion **62B** instead of the covered portion **62**, as well as a stepped portion **63**. The protruding portion **61B** has a straight shape with an outer diameter  $\Phi 1B$  constant in whole. The covered portion **62B** has a straight shape with an outer diameter  $\Phi 2$  constant in whole. The outer diameter  $\Phi 1B$  of the protruding portion **61B** corresponds to the “first outer diameter”, and the outer

diameter  $\Phi 2$  of the covered portion **62B** corresponds to the “second outer diameter”. The outer diameter  $\Phi 2$  (second outer diameter) of the covered portion **62B** is smaller than the outer diameter  $\Phi 1B$  (first outer diameter) of the protruding portion **61B**. The stepped portion **63** is provided between the protruding portion **61B** and the covered portion **62B**, where the outer diameter of the coil body **60B** changes from the first outer diameter  $\Phi 1B$  to the second outer diameter  $\Phi 2$ . As illustrated in FIG. 6, the protruding portion **61B** and the covered portion **62B** extend along the longitudinal direction (axial direction) of the guide wire **1B**, and meanwhile the stepped portion **63** extends along the circumferential direction of the guide wire **1B**.

[0063] As illustrated in FIG. 6, since the protruding portion **61B** has a constant outer diameter  $\Phi 1B$ , the outer diameter of the proximal end of the protruding portion **61B** is also  $\Phi 1B$ . In the guide wire **1B** according to the third embodiment, the outer diameter **1B** of the proximal end of the protruding portion **61B** is equal to the outer diameter  $\Phi 10$  of the distal ends of the insulating tubes **10**, **20**, and **30** (specifically, the outer diameter  $\Phi 10$  of the distal end **11a** of the first tube **10**). The first fixation portion **71B** is provided on the distal end portion **11** of the first tube **10** to fix the distal end portion **11** of the first tube **10** and the stepped portion **63** of the coil body **60B**.

[0064] As described above, the configuration of the coil body **60B** can be variously modified, and the coil body **60B** may be configured such that the protruding portion **61B** and the covered portion **62B** each have a constant outer diameter, and a stepped portion **63** is provided between the protruding portion **61B** and the covered portion **62B**. Also, the guide wire **1B** according to the third embodiment as described above can exhibit the same effect as in the aforementioned first embodiment.

[0065] According to the configuration of the third embodiment, the protruding portion **61B** has a constant first outer diameter  $\Phi 1B$ , and the first outer diameter  $\Phi 1B$  of the protruding portion **61B** is larger than the second outer diameter  $\Phi 2$  of the covered portion **62B**. Thereby, the surface area of the protruding portion **61B** can be increased, i.e. the surface area of the electrode portion **EL** can be increased. As a result, the usability and safeness of the guide wire **1B** can be further improved. In the guide wire **1B** according to the third embodiment, since the outer diameter  $\Phi 1B$  of the proximal end of the protruding portion **61B** is equal to the outer diameter  $\Phi 10$  of the distal end **11a** of the first tube **10**, the outer diameter of the entire distal end side of the guide wire **1B** (specifically, the electrode portion **EL** and the first tube **10**) can be made constant. As a result, the guide wire **1B** can be prevented from being caught in the blood vessel or caught by other devices used in combination. Furthermore, fracture of the coil body **60B** due to the step on the outer surface of the guide wire **1B** on the distal end side can be suppressed.

#### Fourth Embodiment

[0066] FIG. 7 is an enlarged sectional view illustrating a part of a guide wire **1C** on the distal end side, according to the fourth embodiment. The guide wire **1C** according to the fourth embodiment includes a coil body **60C** instead of the coil body **60** and a distal tip **80C** instead of the distal tip **80** in the configuration described in the first embodiment.

[0067] The coil body **60C** includes a protruding portion **61C** instead of the protruding portion **61**. The protruding portion **61C** has, in whole, a tapered shape with a diameter gradually decreasing from the distal end toward the proximal end. An outer diameter  $\Phi 11$  of the distal end of the protruding portion **61C** is larger than an outer diameter  $\Phi 12$  of the proximal end of the protruding portion **61C**. The outer diameter  $\Phi 11$  of the distal end of the protruding portion **61C** is larger than the outer diameter  $\Phi 10$  of the distal ends of the insulating tubes **10**, **20**, and **30** (specifically, the outer diameter  $\Phi 10$  of the distal end **11a** of the first tube **10**). On the other hand, the outer diameter  $\Phi 12$  of the proximal end of the protruding portion **61C** is equal to the outer diameter  $\Phi 10$  of the distal ends of the insulating tubes **10**, **20**, and **30** (specifically, the outer diameter  $\Phi 10$  of the distal end **11a** of the first tube **10**). The distal tip **80C** has the maximum outer diameter (i.e. the outer diameter of the proximal end portion of the distal tip **80C**) substantially equal to the outer diameter  $\Phi 11$  of the distal end of the protruding portion **61C**.

[0068] As described above, the configuration of the coil body **60C** can be variously modified, and the coil body **60C** may be configured such that the protruding portion **61C** and the distal tip **80C** (electrode portion EL) have an outer diameter larger than the outer diameter  $\Phi 10$  of the first tube **10**. Also, the guide wire **1C** according to the fourth embodiment as described above can exhibit the same effect as in the aforementioned first embodiment.

[0069] According to the configuration of the fourth embodiment, since the protruding portion **61C** has a tapered shape with an outer diameter gradually decreasing from the distal end toward the proximal end and the outer diameter  $\Phi 11$  of the distal end of the protruding portion **61C** is larger than the outer diameter  $\Phi 10$  of the distal end **11a** of the first tube **10**, the surface area of the protruding portion **61C** can be increased, i.e. the surface area of the electrode portion EL can be increased. As a result, the usability and safeness of the guide wire **1C** can be further improved. According to the configuration of the fourth embodiment, since the outer diameter  $\Phi 12$  of the proximal end of the protruding portion **61C** is equal to the outer diameter  $\Phi 10$  of the distal end **11a** of the first tube **10**, there is no step on the outer surface of the guide wire **1C** on the distal end side, and the guide wire **1C** can be prevented from being caught in the blood vessel or caught by other devices used in combination. Furthermore, fracture of the coil body **60C** due to the step on the outer surface of the guide wire **1C** on the distal end side can be suppressed.

#### Fifth Embodiment

[0070] FIG. **8** is an explanatory view illustrating a sectional configuration of a guide wire **1D** according to the fifth embodiment. The guide wire **1D** according to the fifth embodiment includes a first tube **10D** instead of the first tube **10**, a second tube **20D** instead of the second tube **20**, and a third tube **30D** instead of the third tube **30** in the configuration described in the first embodiment.

[0071] The first tube **10D** is fixed in a state that the inner peripheral surface **13** of the first tube **10D** is in contact with the outer peripheral surface of the covered portion **62** of the coil body **60**.

Similarly, the second tube **20D** is fixed in a state that the inner peripheral surface **23** of the second tube **20D** is in contact with the outer peripheral surface of the large diameter portion **54** of the core shaft **50**. The third tube **30D** is provided between the first tube **10D** and the second tube **20D** and fixed to the first tube **10D** and the second tube **20D**. As illustrated in FIG. **8**, in the guide wire **1D**, the first tube **10D** and the covered portion **62** are in contact with each other, and the second tube **20D** and the large diameter portion **54** are in contact with each other, and therefore the gas layers **41** and **42** described in the first embodiment are not formed.

[0072] As described above, the configuration of the guide wire **1D** can be variously modified, and the gas layers need not be formed inside the insulating tubes **10**, **20**, and **30**. Also, the guide wire **1D** according to the fifth embodiment as described above can exhibit the same effect as in the aforementioned first embodiment. In the guide wire **1D** according to the fifth embodiment, the diameter of the guide wire **1D** can be decreased.

#### Sixth Embodiment

[0073] FIG. **9** is an explanatory view illustrating a sectional configuration of a guide wire **1E** according to the sixth embodiment. The guide wire **1E** according to the sixth embodiment includes a single insulating tube **10E** instead of the insulating tubes **10**, **20**, and **30** in the configuration described in the first embodiment. The insulating tube **10E** is a hollow cylindrical tubular body made of an insulating resin, which extends from the proximal end side with respect to the distal tip **80** and the protruding portion **61** to the proximal end portion of the large diameter portion **54**.

[0074] As described above, the configuration of the guide wire **1E** can be variously modified, and the guide wire **1E** may be insulated by the single insulating tube **10E**. In this case, the second fixation portion **72** and the third fixation portion **73** may be omitted. Although a case using the single insulating tube **10E** has been described as an example in FIG. **9**, the guide wire **1E** may be insulated using two or four or more tubes. Also, the guide wire **1E** according to the sixth embodiment as described above can exhibit the same effect as in the aforementioned first embodiment.

[0075] The disclosed embodiments have an object to improve the flexibility of a distal end portion of a guide wire having an electrode portion on its distal end. [0076] (1) According to an aspect of the disclosed embodiments, a guide wire is provided. This guide wire includes a conductive core shaft, a conductive coil body surrounding a distal end portion of the core shaft, a conductive distal tip joined to a distal end of the core shaft and a distal end of the coil body, and an insulating tube made of a resin and covering a proximal end portion the coil body and a proximal end portion of the core shaft. The coil body has a covered portion covered by the insulating tube, and a protruding portion protruding from a distal end of the insulating tube toward the distal tip. The protruding portion of the coil body and the distal tip constitute an electrode portion of the guide wire for plasma ablation.

[0077] According to this configuration, the coil body has a distal end protruding portion protruding from the distal end of the insulating tube toward the distal tip, and the protruding portion of the coil body and the distal tip constitute the electrode portion for plasma ablation. That means, according to this configuration, since a surface area of the electrode portion is a sum of a surface area of the coil body protruding portion and a surface area of the distal tip (i.e. the surface area of the coil body protruding portion can be added to the surface area of the electrode portion), the surface area of the electrode portion can be easily increased compared to the conventional configurations with an electrode portion consisting only of a distal tip. If the electrode portion consists only of a distal tip, the distal tip needs to be enlarged for increasing the surface area of the electrode portion, which may compromise the flexibility of the distal end portion of the guide wire. According to this configuration, since the surface area of the electrode portion is a sum of the surface area of the coil body protruding portion and the surface area of the distal tip, the distal tip need not be excessively enlarged, and there is no risk of compromising the flexibility of the distal end portion of the guide wire. Furthermore, the protruding portion has a coil shape, and therefore, even if the distal end portion (electrode portion) of the guide wire collides with a blood vessel wall, the impact can be relieved by the protruding portion, and damage to the blood vessel wall can be suppressed. As a result, in a guide wire having an electrode portion on its distal end, the flexibility of the distal end portion and the safeness of the procedure can be improved. [0078] (2) The guide wire according to the above aspect may be configured such that the protruding portion has a distal end straight portion having a constant outer diameter, and a proximal end tapered portion having an outer diameter gradually decreasing from the distal end of the tapered portion to the proximal end of the tapered portion.

[0079] According to this configuration, since the protruding portion of the coil body has the distal end straight portion having a constant outer diameter, and the proximal end tapered portion having an outer diameter gradually decreasing from the distal end of the coil body toward the proximal end of the coil body, the surface area of the protruding portion of the coil body, i.e. the surface area of the electrode portion can be increased. Since the increase in the surface area of the electrode portion can also contribute to improvement in visibility of the electrode portion in an X-ray image (angiogram), the usability of the guide wire can be improved. Also, the increase in the surface area of the electrode portion leads to reduction in the risk of blood vessel perforation, and therefore the safeness of the guide wire can be improved. [0080] (3) The guide wire according to the above aspects may be configured such that the outer diameter of the proximal end of the straight portion of the protruding portion of the coil body is equal to the outer diameter of the distal end of the insulating tube.

[0081] According to this configuration, since the outer diameter of the proximal end of the straight portion of the protruding portion of the coil body is equal to the outer diameter of the distal end of the insulating tube, the outer diameter of the entire distal end side of the guide wire (specifically, the electrode portion of the guide wire, excluding the tapered portion of the protruding portion of the coil body, and the insulating tube) can be made constant. As a result, the guide wire can be prevented from being caught in the blood vessel or caught by other devices used in combination.

[0082] (4) The guide wire according to the above aspects may be configured such that the protruding portion of the coil body has a constant first outer diameter, the covered portion of the coil body has a constant second outer diameter smaller than the first outer diameter, and the coil body includes, between the protruding portion of the coil body and the covered portion of the coil body, a stepped portion of the coil body where an outer diameter of the coil body changes from the first outer diameter of the protruding portion of the coil body to the second outer diameter of the covered portion of the coil body.

[0083] According to this configuration, since the protruding portion of the coil body has a constant first outer diameter and the first outer diameter of the protruding portion of the coil body is larger than the second outer diameter of the covered portion of the coil body, the surface area of the protruding portion of the coil body can be increased, i.e. the surface area of the electrode portion of the guide wire can be increased. As a result, the usability and the safeness of the guide wire can be further improved. [0084] (5) The guide wire according to the above aspects may be configured

such that the protruding portion of the coil body has a tapered shape with an outer diameter gradually decreasing from the distal end of the protruding portion of the coil body toward the proximal end of the protruding portion of the coil body, and the outer diameter of the distal end of the protruding portion of the coil body is larger than an outer diameter of the distal end of the insulating tube.

[0085] According to this configuration, since the protruding portion of the coil body has a tapered shape with an outer diameter gradually decreasing from the distal end of the protruding portion of the coil body toward the proximal end of the protruding portion of the coil body and the outer diameter of the distal end of the protruding portion of the coil body is larger than the outer diameter of the distal end of the insulating tube, the surface area of the protruding portion of the coil body can be increased, i.e. the surface area of the electrode portion of the guide wire can be increased. As a result, the usability and the safeness of the guide wire can be further improved. [0086] (6) The guide wire according to the above aspects may be configured such that the outer diameter of the proximal end of the protruding portion of the coil body is equal to the outer diameter of the distal end of the insulating tube.

[0087] According to this configuration, since the outer diameter of the proximal end of the protruding portion of the coil body is equal to the outer diameter of the distal end of the insulating tube, the outer diameter of the entire distal end side of the guide wire (specifically, the electrode portion of the guide wire and the insulating tube) can be made constant. As a result, the guide wire can be prevented from being caught in the blood vessel or caught by other devices used in combination. [0088] (7) The guide wire according to the above aspects may be configured such that the electrode portion of the guide wire is a return electrode.

[0089] According to this configuration, since the electrode portion of the guide wire is a return electrode, the guide wire can be configured as a so-called return guide wire that is used in combination with a plasma guide wire. [0090] (8) According to an aspect of the disclosed embodiments, a medical system is provided. This medical system includes the guide wire according to the above aspects, a plasma guide wire having a distal end electrode, and a high frequency generator, in which, when a high frequency wave is applied to the guide wire and the plasma guide wire from the high frequency generator, the electrode portion of the guide wire serves as a return electrode that generates a plasma on the distal end electrode of the plasma guide wire.

[0091] According to this configuration, it is possible to provide a medical system including a plasma guide wire and a guide wire (so-called return guide wire) used in combination with the plasma guide wire.

[0092] The disclosed embodiments can be embodied according to various aspects, and, for example, in a form of a guide wire, a plasma guide wire, a medical system including the guide wire and the plasma guide wire, and a production method therefor.

MODIFICATIONS OF EMBODIMENTS

[0093] The disclosed embodiments are not limited to the above-described embodiments and may be implemented in various modes without departing from the gist thereof, and for example, the following modifications are also possible.

#### Modification 1

[0094] Examples of the configuration of the medical system **1000** have been described in the first to sixth embodiments above. However, the configuration of the medical system **1000** can be variously modified. For example, the configuration of the plasma guide wire **100** described in FIG. **1** is merely an example, and variously modified. For example, instead of the first to third tubes **110** to **130**, one or any number of tubes may be used, and the gas layers **141** to **143** may be omitted. In FIG. **1** and FIG. **4**, a case using the guide wires **1** and **1A** to **1E** as so-called return guide wires has been described as an example. However, the guide wires **1** and **1A** to **1E** may be used as plasma guide wires that generate a plasma on the electrode portion EL. In other words, the guide wires **1** and **1A** to **1E** may be configured to play the role of the plasma guide wire **100** described in FIG. **1** and FIG. **4**.

#### Modification 2

[0095] In the first to sixth embodiments, examples of the configurations of the guide wires **1** and **1A** to **1E** have been described. However, the configurations of the guide wires **1** and **1A** to **1E** can be variously modified. For example, the first tube **10**, the second tube **20**, and the third tube **30** may be integrally formed. For example, the core shaft **50** is not limited to the above-described shape but may have any shape. For example, at least a part of the small diameter portion **51**, the first tapered portion **52**, the second tapered portion **53**, the large diameter portion **54**, and the proximal end portion **55** described as examples in the above embodiments may be omitted. For example, the guide wire may include additional configurations not described above. For example, an inner coil body may be provided inside the coil body **60**. For example, a protective member for protecting the first tube **10** from the electric discharge may be provided on the distal end portion of the first tube **10**. For example, a protective member for protecting the joint part may be provided between the first tube **10** and the third tube **30** or between the third tube **30** and the second tube **20**. For example, a color marker for improving visibility in visual observation or a radiopaque marker for improving visibility in X-ray images may be provided on the distal end **11a** of the first tube **10** or at any position.

#### Modification 3

[0096] The configurations of the guide wires **1** and **1A** to **1E** according to the first to sixth embodiments, and the configurations of the guide wires **1** and **1A** to **1E** according to Modifications **1** and **2** above may be combined as appropriate. For example, the guide wire **1D** described in the fifth embodiment may be configured to include the coil bodies **60A**, **B**, and **C** described in any of the second, third, and fourth embodiments. For example, in the guide wire **1E** described in the sixth embodiment may be configured to include the coil bodies **60A**, **B**, and **C** described in any of the second, third, and fourth embodiments.

[0097] Although the present aspects have been explained above based on the embodiments and the modifications, the embodiments of the above-described aspects are intended to facilitate understanding of the present aspects and do not limit the present aspects. The present aspects may be modified and improved without departing from the gist and the scope of claims and include equivalents thereof. Unless the technical features are described as essential in the present specification, the technical features may be deleted as appropriate.

#### DESCRIPTION OF REFERENCE NUMERALS

[0098] **1**, **1A** to **1E** . . . Guide wire [0099] **10**, **10D**, **10E** . . . First tube (insulating tube) [0100] **20**, **20D** . . . Second tube (insulating tube) [0101] **30**, **30D** . . . Third tube (insulating tube) [0102] **41**, **42**, **43** . . . Gas layer [0103] **50** . . . Core shaft [0104] **51** . . . Small diameter portion [0105] **52** . . . First tapered portion [0106] **53** . . . Second tapered portion [0107] **54** . . . Large diameter portion [0108] **55** . . . Proximal end portion [0109] **60**, **60A** to **60C** . . . Coil body [0110] **61**, **61A** to **61C** . . .



Protruding portion [0111] **62**, **62B** . . . Coated portion [0112] **63** . . . Stepped portion [0113] **70** . . . Coil fixation portion [0114] **71**, **71B** . . . First fixation portion [0115] **72** . . . Second fixation portion [0116] **73** . . . Third fixation portion [0117] **74** . . . Fourth fixation portion [0118] **80**, **80A**, **80C** . . . Distal tip [0119] **82** . . . Distal end side joint part [0120] **83** . . . Proximal end side joint part [0121] **100** . . . Plasma guide wire [0122] **110** . . . First tube [0123] **120** . . . Second tube [0124] **130** . . . Third tube [0125] **141**, **142**, **143** . . . Gas layer [0126] **150** . . . Core shaft [0127] **160** . . . Coil body [0128] **170** . . . Coil fixation portion [0129] **171** . . . First fixation portion [0130] **172** . . . Second fixation portion [0131] **173** . . . Third fixation portion [0132] **174** . . . Fourth fixation portion [0133] **180** . . . Distal tip [0134] **200** . . . RF generator [0135] **210** . . . First terminal [0136] **211** . . . First cable [0137] **220** . . . Second terminal [0138] **221** . . . Second cable [0139] **300** . . . Catheter [0140] **301** . . . First shaft [0141] **302** . . . Second shaft [0142] **303** . . . Distal tip [0143] **611** Straight portion [0144] **612** . . . Tapered portion [0145] **1000** . . . Medical system

## Claims

1. A guide wire comprising: a conductive core shaft; a conductive coil body surrounding a distal end portion of the core shaft; a conductive distal tip joined to a distal end of the core shaft and a distal end of the coil body; and an insulating tube made of a resin and covering a proximal end portion of the coil body and a proximal end portion of the core shaft, the coil body having a covered portion covered by the insulating tube and a protruding portion protruding from a distal end of the insulating tube toward the distal tip, wherein the protruding portion of the coil body and the distal tip constitute an electrode portion of the guide wire for plasma ablation.
2. The guide wire according to claim 1, wherein the protruding portion of the coil body has: a straight portion having a constant outer diameter, and a tapered portion disposed on a proximal end side with respect to the straight portion and having an outer diameter gradually decreasing from a distal end of the tapered portion to a proximal end of the tapered portion.
3. The guide wire according to claim 2, wherein the outer diameter of a proximal end of the straight portion is equal to an outer diameter of the distal end of the insulating tube.
4. The guide wire according to claim 1, wherein the protruding portion has a constant first outer diameter, the covered portion has a constant second outer diameter smaller than the first outer diameter, and the coil body comprises, between the protruding portion and the covered portion, a stepped portion where an outer diameter of the coil body changes from the first outer diameter to the second outer diameter.
5. The guide wire according to claim 1, wherein the protruding portion has a tapered shape with an outer diameter gradually decreasing from a distal end toward a proximal end, and the outer diameter of the distal end of the protruding portion is larger than an outer diameter of the distal end of the insulating tube.
6. The guide wire according to claim 4, wherein the outer diameter of the proximal end of the protruding portion is equal to the outer diameter of the distal end of the insulating tube.
7. The guide wire according to claim 1, wherein the electrode portion is a return electrode.
8. The guide wire according to claim 5, wherein the outer diameter of the proximal end of the protruding portion is equal to the outer diameter of the distal end of the insulating tube.
9. The guide wire according to claim 2, wherein the electrode portion is a return electrode.
10. The guide wire according to claim 3, wherein the electrode portion is a return electrode.
11. The guide wire according to claim 4, wherein the electrode portion is a return electrode.
12. The guide wire according to claim 5, wherein the electrode portion is a return electrode.
13. The guide wire according to claim 6, wherein the electrode portion is a return electrode.
14. A medical system comprising: the guide wire according to claim 1; and a plasma guide wire having a distal end electrode, wherein when a high frequency wave is applied to the guide wire and the plasma guide wire from a high frequency generator, the electrode portion of the guide wire

serves as a return electrode that generates a plasma on the distal end electrode of the plasma guide wire.

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